

Machine learning and deep learning in the energy sector

Applications, technical foundations and product view

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Abstract

This document describes four application areas of machine learning (ML) and deep learning (DL) in the power sector: (1) forecasting of load, renewable generation and price; (2) grid operation and optimization; (3) fault detection, protection and predictive maintenance; and (4) distribution and the end consumer. For each area it presents the business problem, the mathematical formulation, the model families, the required data, the evaluation metrics and the practical challenges. Each section ends with a product view: who the user is, which problem is addressed, the minimum viable scope, the reference architecture, the success metrics, the business model and the risks. The Brazilian regulatory and operational context (ONS, CCEE, ANEEL, distribution utilities) is used as the reference where applicable.

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1 Introduction

The power sector is undergoing three simultaneous transformations that widen the scope for ML/DL:

1. **Decarbonization:** the growing share of wind and solar sources, which are variable and non-dispatchable, shifts the central problem of operation from “serving a predictable load with controllable generation” to “balancing uncertainties on both sides”.
2. **Decentralization:** distributed micro and mini generation (MMGD, from the Portuguese *micro e minigeração distribuída*), batteries and electric vehicles turn the consumer into an active agent and make the distribution grid bidirectional.
3. **Digitalization:** smart meters, phasor measurement units (PMUs), sensors on assets, drones and satellite imagery generate volumes of data that classical methods do not exploit well.

Despite the large volume of research, recent reviews converge on one diagnosis: there are many pilots and few systems at scale, and the main barrier is not model accuracy but integration with legacy systems, reliability and acceptance by operators [1, 2]. For this reason, each section of this document treats the model and the product as equally important problems.

2 Forecasting: load, renewable generation and price

2.1 Context and importance

Forecasting is one of the most mature applications of ML in the sector. Operational and commercial decisions depend on an estimate of the future:

- the **system operator** (in Brazil, ONS) needs to forecast load and renewable generation to schedule dispatch, size reserves and assess security;
- **generators and trading companies** need to forecast production and price to contract, hedge and avoid exposure to the short-term market;
- **distribution utilities** need to forecast demand for energy purchases in auctions, expansion planning and management of transformer loading.

A frequently cited case of value generated is that of DeepMind with Google: a neural network trained on weather forecasts and turbine history started to forecast the generation of 700 MW of wind farms 36 hours ahead. With that forecast, Google started to offer the energy in the day-ahead market, and the company reports an increase of about 20% in the value of the wind energy [3].

2.2 Forecast horizons

Table 1
Forecast horizons and their uses.

Horizon	Typical range	Main use
Very short term	seconds to 1 h	frequency control, ramps, intra-hour dispatch, inverter control
Short term	1 h to 7 days	daily operation scheduling, day-ahead market, unit commitment
Medium term	weeks to months	operation planning (PMO), maintenance, reservoir management, contracting
Long term	years	generation and transmission expansion, auctions, tariff planning

Each horizon has different data, models and metrics. At very short horizons, local high-resolution data predominate (measurements, sky cameras, satellite images). In the short term, numerical

weather prediction (NWP) is the main explanatory variable. In the medium and long term, macroeconomic and climate scenarios enter, and the problem is closer to structural modeling than to pure ML.

2.3 Problem formulation

Let y_t be the target variable (load, generated power or price) at time t , $\mathbf{x}_t$ a vector of observed covariates (temperature, measured irradiance, etc.) and $\mathbf{z}_{t+h}$ covariates known or forecast for the future (calendar, holidays, NWP). The point forecast with horizon h and context window L is

$$\hat{y}_{t+h} = f_{\theta}(y_{t-L+1:t}, \mathbf{x}_{t-L+1:t}, \mathbf{z}_{t+1:t+h}). \quad (1)$$

In practice, the decision rarely depends on the expected value alone. Sizing reserve, bidding in a market or deciding whether to start a thermal plant requires knowing the *distribution* of the future. For this reason the trend is toward **probabilistic forecasting**, usually through quantiles q_{τ} trained with the pinball loss:

$$\mathcal{L}_{\tau}(y, q_{\tau}) = \max(\tau(y - q_{\tau}), (\tau - 1)(y - q_{\tau})), \quad \tau \in (0, 1). \quad (2)$$

The quality of the full distribution is measured by the Continuous Ranked Probability Score (CRPS), where F is the forecast distribution:

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{I}\{z \geq y\})^2 dz. \quad (3)$$

The two measures are related: the CRPS is the pinball loss integrated over all quantiles,

$$\text{CRPS}(F, y) = 2 \int_0^1 \mathcal{L}_{\tau}(y, F^{-1}(\tau)) d\tau, \quad (4)$$

and it reduces to the absolute error when the forecast is a point forecast.

The quantile that matters depends on the decision. If each MWh under-forecast costs c_- and each MWh over-forecast costs c_+ , the forecast that minimizes the expected cost is the quantile

$$\tau^* = \frac{c_-}{c_- + c_+}, \quad (5)$$

which coincides with the median only when the two costs are equal. A fixed set of quantiles (P10, P50, P90) serves the decision only if τ^* is one of them.

2.4 Load forecasting

Load drivers. Aggregate electrical load has strong and relatively stable patterns:

- **multiple seasonality:** daily, weekly and annual;
- **calendar effects:** holidays, bridge days, daylight saving time (where it exists), large events;
- **weather:** temperature is the main variable (a nonlinear, U-shaped relationship, because of air conditioning and heating), followed by humidity, apparent temperature and cloud cover;
- **economy:** industrial and commercial activity, shocks such as the 2020 pandemic.

Gross load versus net load. With the expansion of solar MMGD, what the operator observes at the connection points is the *net load*: consumption minus behind-the-meter generation. On sunny days, the net load drops at midday and rises abruptly in the late afternoon, forming the so-called “duck curve”. Forecasting the net load directly mixes two phenomena with different dynamics. For this reason, the common practice is to decompose:

$$\text{net load}_t = \text{gross load}_t - \text{MMGD}_t, \quad (6)$$

forecasting each term with its own covariates. For this reason, ONS maintains a dedicated system for forecasting photovoltaic MMGD generation, with a 72-hour horizon [5].

The decomposition has two conditions. The first is one of identification: only the net load is measured. Behind-the-meter MMGD has to be estimated (registered installed capacity and irradiance), and the gross load is reconstructed from that estimate, so the error of the estimate enters both terms. The second is one of variance: with e_b and e_m the forecast errors of the gross load and of MMGD,

$$\text{Var}(e_b - e_m) = \text{Var}(e_b) + \text{Var}(e_m) - 2 \text{Cov}(e_b, e_m), \quad (7)$$

and the decomposition outperforms the direct forecast only if this variance is smaller than the error variance of the direct model. The comparison is made by backtesting.

Hierarchical forecasting. Load exists at several levels (subsystem, area, substation, feeder, transformer, consumer). Independent forecasts at each level do not add up coherently. **Hierarchical reconciliation** techniques (such as MinT) adjust the forecasts so that the levels are consistent with each other, and they tend to improve accuracy at all levels.

2.5 Wind generation forecasting

The power extracted from the wind grows with the cube of the speed:

$$P = \frac{1}{2} \rho A C_p(\lambda, \beta) v^3, \quad (8)$$

where ρ is the air density, A the swept area, C_p the power coefficient and v the wind speed. With constant C_p , the relative error triples: $\delta P/P \approx 3 \delta v/v$, and an error of 10% in the wind corresponds to 33% in power ($1.1^3 = 1.331$). The relationship holds in the cubic region of the power curve; above rated speed the power is limited and the sensitivity to wind falls to zero.

The cube also biases the conversion. For wind with mean $\bar{v}$ and turbulence intensity $I = \sigma_v/\bar{v}$,

$$\mathbb{E}[v^3] \approx \bar{v}^3 (1 + 3I^2), \quad (9)$$

so applying the power curve to the mean wind forecast underestimates the mean power: with $I = 0.15$, $\mathbb{E}[v^3]$ exceeds $\bar{v}^3$ by 6.75%. Near rated speed the curve is concave and the bias changes sign.

The typical architecture of short-term forecasting has two stages:

1. **Weather forecast:** global or regional NWP models provide wind at several heights, direction, temperature and pressure. AI-based weather models (such as GraphCast, GenCast and Aurora) have come to compete with physical NWP on several metrics, at a much lower computational cost.
2. **Conversion model:** an ML model (usually gradient boosting or neural networks) learns the relationship between the forecast weather and the measured power of the farm. This model corrects NWP biases, wake effects, unavailability and losses.

ONS developed its own wind forecasting model, aimed at the Northeast and South regions, to reduce the need for power reserve [6].

NOTE Caution with curtailment. When the operator restricts the generation of a farm (in Brazil, the so-called *constrained-off*, which has become relevant for wind and solar plants in the Northeast), the measured power no longer reflects the available resource. A model trained on these data without filtering them underestimates generation. The periods of restriction need to be flagged and excluded, or the *available* power modeled instead of the *delivered* power.

2.6 Solar generation forecasting

The irradiance that reaches the surface is dominated by a deterministic component (the position of the sun) and a stochastic component (clouds and aerosols). The standard normalization is the **clear-sky index**:

$$k_t^{\text{cs}} = \frac{\text{GHI}_t}{\text{GHI}_t^{\text{cs}}}, \quad (10)$$

where GHI^{cs} is the global horizontal irradiance computed by a clear-sky model. Forecasting k_t^{cs} instead of GHI removes the astronomical seasonality and restricts the problem to the stochastic component: the clouds. The clear-sky index is not the clearness index, which divides GHI by the extraterrestrial irradiance and is usually written k_t .

The denominator tends to zero at sunrise and sunset, and the ratio amplifies the measurement error at those times. For this reason the index is computed and evaluated only above a minimum solar elevation (of the order of 5°). The index also exceeds 1 in episodes of reflection at cloud edges.

The data sources and models vary with the horizon:

- **minutes:** all-sky imagers and CNNs that estimate cloud motion;
- **up to 6 hours:** geostationary satellite images (in Brazil, GOES-East) and video/spatiotemporal models (ConvLSTM, U-Net);
- **day ahead and beyond:** NWP as input to ML models.

The conversion from irradiance to power involves module temperature (efficiency falls with heat), orientation, shading, soiling and the inverter power limit (clipping). A Brazilian study uses LSTM for hourly forecasting of photovoltaic generation [7].

2.7 Price forecasting

In bid-based markets (Europe, USA), deep networks are among the methods with the lowest reported error for day-ahead price forecasting. Brazil is different: the Settlement Price for Differences (PLD, *Preço de Liquidação das Diferenças*) does not result from bids; it is computed by CCEE from the chain of hydrothermal optimization models (NEWAVE, DECOMP and DESSEM, developed by CEPEL). Since 2021 the PLD has been hourly and computed by DESSEM.

This opens two ML strategies:

1. **Direct PLD forecasting** from variables such as inflows, storage, load and forecast renewable generation;
2. **Emulation of the official models** (surrogate modeling): training a network to approximate the output of DESSEM/DECOMP as a function of the inputs, which allows thousands of scenarios to be run in seconds. It is useful for the risk analysis of trading companies.

2.8 Model families

Table 2
Model families for forecasting in energy.

Family	Strengths	Limitations
Statistical (ARIMA, ETS, regression)	interpretable, low cost, suitable as baselines	capture nonlinearities and many covariates poorly
Gradient boosting (XGBoost, LightGBM)	good performance on tabular data with rich covariates; widely used in industry	requires feature engineering (lags, windows)
RNN (LSTM, GRU)	model temporal dependence directly	slower training; do not consistently outperform GBM
Temporal CNNs (TCN)	parallelizable, long receptive field	less used with future covariates
N-BEATS, N-HiTS	deep MLPs with interpretable decomposition	univariate in the original form
Transformers (TFT, PatchTST)	heterogeneous covariates, multiple horizons, quantiles (TFT) [8]	require more data and tuning
Foundation models (Chronos, TimesFM, Moirai)	zero-shot forecasting without local training [9, 10]	limited covariate support; inference cost

Time series foundation models. These are models pretrained on large collections of heterogeneous time series, able to forecast unseen series without retraining. The line started with TimeGPT (2023) and continued with Lag-Llama, Chronos, TimesFM and, more recently, Sundial and Kairos [11]. Results from 2025–2026 in energy:

- in load benchmarks at several grid levels, time series Transformers outperformed the established methods [12];
- foundation models run on consumer hardware with good zero-shot performance, but with sensitivity to context length and to distribution shift [13];
- the incorporation of covariates (weather, calendar) improves the performance of the models and allows the forecasts to be explained [14];
- at low voltage, where the series are noisy and short, the models were applied to probabilistic peak forecasting [15];
- energy-specific models, such as GridFM, jointly forecast load, price, emissions and renewables, incorporating physical constraints [16].

The main practical benefit is the **cold start**: new farms, new feeders or new customers receive forecasts on the first day, before there is history to train a dedicated model.

2.9 Evaluation

Metrics.

- **MAE, RMSE**: in the unit of the variable; the RMSE penalizes large errors more, which matters for ramps.
- **MAPE**: popular for load, but *unsuitable for solar*, which is zero at night (division by zero) and has small values at the start and end of the day. For renewables, the error normalized by installed capacity is used (nMAE, nRMSE).
- **Skill score**: relative gain over a naive reference (persistence or climatology), $SS = 1 - RMSE_{\text{model}}/RMSE_{\text{ref}}$. Without this comparison, absolute error numbers are not very informative.
- **Probabilistic**: pinball loss, CRPS, coverage and width of the intervals (calibration and sharpness).
- **Value metrics**: avoided imbalance cost, saved reserve, additional revenue. These are the metrics that matter to the business.

Validation. Two common errors artificially inflate the results:

1. **Random cross-validation** on time series leaks the future into training. The appropriate procedure is backtesting with a rolling origin, which respects temporal order.
2. **Covariate leakage**: using the *observed* weather in place of the *forecast* available at decision time. In production only the forecast exists, issued at a specific time. Training needs to reproduce what would have been available at each moment.

2.10 Challenges

- **Extreme events**: heat waves, storms and blackouts are rare in the history and are the ones with the largest consequences.
- **Structural changes**: growth of MMGD, electrification, changes in habits. Models need drift monitoring and continuous retraining.
- **Data quality**: missing measurements, telemetry errors, unflagged curtailment.
- **User trust**: the operator needs to understand why the forecast changed. Explainability (covariate importance, decomposition) is a requirement.

2.11 Product view: Forecasting platform for renewables and load

Problem. Wind and solar generators are exposed to imbalance penalties, curtailment and the short-term price. Trading companies need reliable forecasts to contract. Distribution utilities and large consumers have an increasingly uncertain net load because of MMGD. Many of these agents use spreadsheets, generic forecasts from foreign vendors or persistence.

Users and personas.

- **Commercial operations analyst of a generator:** needs the generation forecast per farm for scheduling and for declarations to the operator.
- **Trader at a trading company:** needs forecasts of aggregate generation, load and PLD, with uncertainty, to decide positions.
- **Distribution utility planner:** needs the load per substation/feeder and the MMGD per region.

Value proposition. Probabilistic forecasts per asset, with horizons from minutes to 15 days, available through an API and a dashboard, calibrated for the Brazilian context (NWP and satellite over the territory, national calendar, curtailment flagging), with a measured gain over the reference that the customer already uses.

MVP scope.

1. Day-ahead and intraday forecasting of wind and solar generation for farms with available telemetry.
2. Quantiles (P10, P50, P90), not only a point value.
3. Dashboard with a daily comparison of forecast versus actual and skill score against persistence.
4. REST API for integration with the customer's systems.
5. Onboarding of new farms with a zero-shot foundation model, replaced by a fitted model once there is history.

Evolution (roadmap).

- **V2:** nowcasting with satellite images (0–6 h); automatic detection of curtailment and unavailability.
- **V3:** forecasting of PLD and of load per submarket; joint scenarios (generation, load, price) for risk analysis.
- **V4:** optimization of bids and of battery dispatch coupled to the forecasts (from forecast to decision).

Reference architecture.

- **Ingestion:** NWP (multiple providers), satellite, SCADA telemetry of the farms, open data from ONS and CCEE.
- **Feature store** with temporal versioning (“what was known at each instant”) to avoid leakage.
- **Training** per asset or global (one model for several farms), with automatic selection by backtesting.
- **Inference service** scheduled according to the arrival of each NWP run.
- **Monitoring** of error in production, drift and alerts for missing data.

Success metrics.

- Technical: nMAE and CRPS per asset; skill score over persistence and over the customer's current vendor.
- Business: reduction in imbalance cost (R\$/MWh), additional revenue, analyst hours saved.
- Product: onboarding time for a new farm, API availability, customer retention.

Business model. Subscription per monitored MW or per asset, with tiers (day ahead; intraday; nowcasting; price). A free or low-cost pilot in shadow mode, compared with the customer's current forecast, is the main sales instrument: the gain appears in the customer's own numbers.

Risks.

- Dependence on third-party NWP (cost, availability, version changes that alter the input distribution).
- Access to the customer's telemetry (bureaucracy, information security).
- A market with established international competitors; differentiation needs to come from local accuracy and integration with the Brazilian regulatory context.

3 Grid operation and optimization

3.1 Context

Operating a power system in real time involves repeatedly solving analysis and optimization problems:

- **Power flow:** given the state of generation and load, compute voltages and flows across the whole grid;
- **Contingency analysis:** check whether the system remains secure after the loss of any element ($N - 1$ criterion), which requires thousands of power flows;
- **Optimal power flow (OPF):** choose the lowest-cost dispatch that respects all physical limits;
- **State estimation:** reconstruct the most likely state of the grid from noisy and incomplete measurements;
- **Control:** voltage and reactive power, redispatch, switching (topology), batteries.

With more renewables, these problems need to be solved more often, for more scenarios and with more uncertainty. Classical solvers are reliable, but the computational cost grows rapidly with the size of the grid and the number of scenarios. ML enters here mainly as an **accelerator** and as **decision support**, not as a substitute for the physical methods.

3.2 Formulation: AC power flow

For each bus i of the grid, the active and reactive power injections are

$$P_i = \sum_j |V_i||V_j|(G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \quad (11)$$

$$Q_i = \sum_j |V_i||V_j|(G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}), \quad (12)$$

where V_i is the voltage phasor, $\theta_{ij} = \theta_i - \theta_j$ and $G_{ij} + jB_{ij}$ are the elements of the nodal admittance matrix. The system is nonlinear and is solved by Newton–Raphson.

The OPF adds an objective function and constraints:

$$\begin{aligned}
 & \min_{p,q,|V|,\theta} \sum_g c_g(p_g) \\
 & \text{s.t. } P_i = \sum_{g \in \mathcal{G}_i} p_g - P_i^d, \quad Q_i = \sum_{g \in \mathcal{G}_i} q_g - Q_i^d, \\
 & V_i^{\min} \leq |V_i| \leq V_i^{\max}, \\
 & p_g^{\min} \leq p_g \leq p_g^{\max}, \quad q_g^{\min} \leq q_g \leq q_g^{\max}, \\
 & |S_{ij}| \leq S_{ij}^{\max}, \quad |S_{ji}| \leq S_{ij}^{\max}, \\
 & \theta_{\text{ref}} = 0.
 \end{aligned} \tag{13}$$

Here p_g and q_g are the active and reactive power of generator g , $\mathcal{G}_i$ is the set of generators at bus i , P_i^d and Q_i^d are the load at the bus, and P_i and Q_i are the injections of the power flow equations. The flow limit applies at both terminals of the line, because losses make $|S_{ij}| \neq |S_{ji}|$, and the reference bus fixes the origin of the angles. The problem is nonconvex and, in large systems with security constraints, costly to solve within real-time deadlines.

3.3 Graph neural networks (GNNs)

A power grid can be represented as a graph: buses are nodes, lines and transformers are edges. GNNs process this type of data by message passing:

$$\mathbf{h}_i^{(k+1)} = \phi\left(\mathbf{h}_i^{(k)}, \bigoplus_{j \in \mathcal{N}(i)} \psi(\mathbf{h}_i^{(k)}, \mathbf{h}_j^{(k)}, \mathbf{e}_{ij})\right), \tag{14}$$

where $\mathbf{h}_i$ is the representation of bus i (load, generation, type), $\mathbf{e}_{ij}$ are attributes of the line (impedance, limit) and $\bigoplus$ is a permutation-invariant aggregation (sum, mean). After K layers, the representation of each bus incorporates its K -hop neighborhood.

This structure brings two advantages over MLPs that receive a fixed vector:

1. **Topological generalization:** the same GNN accepts as input the grid without a line or with the topology changed, because the parameters are shared among nodes and edges.
2. **Scalability:** the cost grows with the number of edges, and models trained on smaller grids can be applied to larger grids.

Accepting the input does not guarantee a correct output. Power flow is a global problem: in the linear approximation, the voltages are $\mathbf{V} = \mathbf{Y}^{-1}\mathbf{I}$, and $\mathbf{Y}^{-1}$ is dense even when $\mathbf{Y}$ is sparse, so the voltage at one bus depends on the injections of the whole grid. A GNN with K smaller than the diameter of the graph does not represent this dependence. Two limits to the advantages above follow from this: the outage of a line redistributes flow beyond the K hops, and a larger grid has a larger diameter, which requires more layers or global aggregation. The error under these two conditions is measured by testing with topologies and sizes outside the training set.

Main applications.

- **Power flow surrogate:** predicting voltages and flows directly, in milliseconds, for fast contingency screening. A recent study reports GNNs that scale to larger systems with a computation time lower than that of traditional solvers [17].
- **OPF warm start:** the GNN provides an initial point close to the solution, and the classical solver converges in fewer iterations, which preserves the feasibility guarantee [18].
- **Network reconfiguration:** predicting the topology before the optimization stage, which reduces the combinatorial search space [19].

3.4 Learning to optimize and the feasibility problem

Training a network to map inputs (load, costs) directly to OPF solutions (optimization proxies) has a limitation: the network does not guarantee that the solution respects the constraints. A violation of

a thermal or voltage limit is not a “small error”; it is an unacceptable solution. The main strategies are:

- **Penalization** of the violations in the loss function (low implementation cost, no guarantee);
- **Lagrangian duality** during training, adjusting the weights of the constraints;
- **Correction and completion**: the network predicts part of the variables and the others are obtained by solving the equalities, followed by correction steps for the inequalities (as in DC3 [20]);
- **Projection** of the output onto the feasible set, or use of the prediction only as a warm start for an exact solver.

In industrial practice, the most accepted approach is the last one: ML accelerates, the solver certifies.

3.5 Reinforcement learning (RL)

Sequential control problems, such as deciding switching actions, redispatch or battery cycles over time, can be formulated as Markov decision processes. The agent observes the state s_t (flows, voltages, forecasts), chooses an action a_t and receives a reward r_t (avoided cost, security margin). The objective is

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]. \quad (15)$$

Topology control. The *Learning to Run a Power Network* (L2RPN) competition, organized by RTE (the French operator), popularized the use of RL to keep the grid operating through topology changes in substations, redispatch and curtailment [21]. Agents that combine GNNs with RL (graph RL), sometimes pretrained by imitation learning and then refined with PPO, were proposed to handle topology changes caused by extreme weather or maintenance [22, 23].

Batteries and microgrids. RL is used to decide when to charge and discharge batteries considering price, generation forecast and degradation. PINN-based surrogates allow agents to be trained without interacting with the real system [24].

Safe RL. In physical systems, exploring bad actions has a real cost. The standard formulation is the constrained MDP:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t r_t \right] \quad \text{s.t.} \quad \mathbb{E}_{\pi} \left[\sum_t \gamma^t c_t \right] \leq d, \quad (16)$$

where c_t measures violations (overload, undervoltage). The constraint holds in expectation and with discounting: a policy satisfies it even with occasional overloads, as long as the discounted mean stays below d . A physical limit calls for a per-instant constraint, in chance form,

$$\Pr(c_t > 0) \leq \epsilon \quad \text{for all } t, \quad (17)$$

or a constraint on the tail of the cost distribution (CVaR). Other techniques include shielding (a module checks and blocks dangerous actions before execution) and offline training from historical operator data [25].

3.6 Physics-informed neural networks (PINNs)

PINNs incorporate physical equations into the loss function:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{physics}}, \quad (18)$$

where $\mathcal{L}_{\text{physics}}$ penalizes the residual of the equations (for example, of the power flow equations or of the differential equations of machine dynamics) [27]. The reported benefits are a lower need for labeled data, closer adherence to physical laws and better extrapolation. PINNs are used in state estimation, transient stability analysis and as simulators for training RL agents.

3.7 Challenges

- **Certification and liability:** operators and regulators require guarantees; a model that is correct in 99.9% of the cases can still fail in the critical contingency. In a sweep of 5,000 contingencies with independent errors, the probability of at least one error is $1 - 0.999^{5000} \approx 99.3\%$.
- **Generalization:** the real grid changes (new lines, new plants), and models trained on past scenarios may not cover future situations [26].
- **Integration with legacy EMS:** energy management systems are critical, closed and have long update cycles.
- **Data:** detailed grid models and operational data are sensitive; research depends heavily on synthetic test systems (IEEE, PGLib).
- **Human factor:** full autonomy is not realistic in the short term. The viable role is that of a copilot for the operator.

3.8 Product view: Grid operation copilot

Problem. Control center operators (in transmission or distribution) need to make fast decisions in the face of a growing number of scenarios: renewable variability, curtailment, contingencies, transmission constraints. The analysis tools are slow for exploring many alternatives in real time, and the experience of senior operators is not codified.

Users and personas.

- **Real-time operator** of a transmission company or of the distribution operation center (COD);
- **Electrical studies engineer** who prepares the operation of the following day;
- **Operator of renewable farms and batteries** who needs to decide dispatch and storage.

Value proposition. Contingency screening and recommendation of corrective actions in seconds, with each recommendation verified by a physical solver before it is presented. The operator remains in command; the system increases the number of scenarios the operator can evaluate.

MVP scope.

1. GNN-based power flow surrogate, trained on simulations of the customer's grid model.
2. Fast screening of $N - 1$ contingencies for the next intervals, using load and renewable forecasts, with the critical cases confirmed by a full power flow.
3. "Hot spot" dashboard: which lines and buses are at risk and when.
4. Operation in **shadow mode**: the system runs in parallel, without acting, and the recommendations are recorded against the actual decisions.

Evolution (roadmap).

- **V2:** recommendation of corrective actions (redispatch, switching), ranked and verified by the solver.
- **V3:** RL agent trained on historical decisions and in simulation, suggesting sequences of actions with security constraints.
- **V4:** coordinated optimization of batteries and curtailment in real time for hybrid farms.

Reference architecture.

- **Digital twin** of the grid (electrical model synchronized with the current topology).
- **Scenario generator** to create synthetic training data from the physical solver.

- **Models:** power flow GNN; contingency ranker; RL policy with a shield.
- **Verification layer:** every output passes through the solver before reaching the operator.
- **Integration** with SCADA/EMS through standard protocols (such as IEC 61850 and CIM), in read-only mode at first.

Success metrics.

- Technical: voltage and flow error of the surrogate; recall of critical contingencies (missing a critical one is the unacceptable error); speed-up relative to the solver.
- Operational: reduction of avoidable curtailment, reduction of violations, response time to events.
- Adoption: percentage of recommendations accepted by the operators.

Business model. Software license per control center, with a service for deploying and calibrating the digital twin. For transmission companies and distribution utilities, R&D projects regulated by ANEEL are an entry route for pilots.

Risks.

- Long sales cycle and risk aversion of the customers.
- Access to the grid model and to operational data.
- Cybersecurity requirements for any integration with control systems.
- Liability in case of a wrong recommendation; the “human in control” design and the physical verification mitigate it but do not eliminate it.

4 Fault detection, protection and predictive maintenance

4.1 Context

Equipment failures cause supply interruptions, damage to costly assets and safety risks. In Brazil, ANEEL measures continuity of supply through collective indicators (DEC and FEC, the equivalent duration and frequency of interruption per consumer unit) and individual ones (DIC, FIC, DMIC), and the violation of the individual limits generates compensation paid to consumers. Reducing the frequency and duration of failures therefore has direct financial value.

Maintenance evolves in four stages:

1. **Corrective:** repair after failure;
2. **Preventive:** periodic interventions by calendar;
3. **Condition-based:** intervene when measured indicators exceed limits;
4. **Predictive:** estimate when the failure is expected to occur and act before it.

ML enables the move from the third to the fourth stage at large scale.

4.2 Fault detection, classification and location

A fault (short circuit) on a line produces voltage and current transients recorded by digital relays and fault recorders. The analysis involves three tasks [28]:

- **Detection:** was there a fault?
- **Classification:** of which type (phase-to-ground, phase-to-phase, three-phase, etc.)?
- **Location:** at what distance from the terminal?

Approaches.

- **Feature extraction + classifier:** wavelet or Fourier transform of the signals, followed by SVM, trees or shallow networks.
- **End-to-end learning:** 1D CNNs on the raw waveforms or 2D CNNs on time-frequency representations (spectrograms, scalograms).
- **PMU data:** synchronized phasor measurements at a high rate allow oscillations and events to be detected across the whole grid; spatiotemporal models (GNN + temporal networks) exploit the topology.
- **Distribution grids:** location is harder because of the branched radial topology and the lower level of instrumentation; ML combines data from smart meters, customer calls and reclosers.

Real fault data are scarce. For this reason, the models are often trained on **electromagnetic transient simulations** (ATP/EMTP, PSCAD), which creates the gap between simulation and reality (sim-to-real), one of the main risks of these applications.

4.3 Predictive maintenance of assets

Power transformers. They are among the most costly and critical assets. **Dissolved gas analysis** (DGA) of the insulating oil is the main diagnostic source: different types of internal fault (partial discharges, arcing, overheating) generate characteristic proportions of hydrogen, methane, ethylene, acetylene and other gases. Classical methods, such as the Duval triangle, are graphical rules; ML allows DGA to be combined with load history, temperature, age and tests to produce a health index and a failure probability.

Wind turbines. The SCADA systems of the turbines record, every 10 minutes, dozens of variables (bearing and gearbox temperatures, power, wind, rotation). A widely used technique is the **normal behavior model**:

1. a model is trained to predict one variable (for example, bearing temperature) from the others, using only healthy periods;
2. in operation, the residual $r_t = y_t - \hat{y}_t$ is monitored;
3. a persistent deviation of the residual indicates degradation, often weeks before the failure.

A “persistent deviation” needs a defined statistic, because the threshold fixes the false alarm rate. A 3σ threshold applied to each sample, with a Gaussian and independent residual, fires on 0.27% of the samples: 0.39 false alarms per day per variable (144 samples of 10 minutes) and about 390 per day in a fleet of 100 turbines with 10 monitored variables. Statistics that accumulate the deviation over time (CUSUM, EWMA) reduce this rate for the same detection power. High-frequency vibration sensors allow a finer diagnosis (bearings, gears) with CNNs on spectra.

Photovoltaic plants. Detection of underperforming strings by comparison with neighboring strings and with the expected irradiance; inverter failures; degradation and soiling (which also guides the cleaning schedule).

Remaining useful life (RUL). When there are enough failures in the history, the time to failure can be estimated with survival models (Cox, random survival forests, survival networks) or direct regression with recurrent networks. The result feeds the optimization of the maintenance plan, which weighs the cost of the intervention, the failure probability and the consequence.

4.4 Computer vision for inspection

- **Transmission and distribution lines:** detection of damaged insulators, missing dampers, corrosion and nests in images from drones and helicopters, with object detectors (YOLO family, DETR).

- **Vegetation:** LiDAR point clouds and satellite images to measure the distance between canopies and conductors and to prioritize pruning; contact with vegetation is one of the main causes of failures in distribution.
- **Thermography:** hot spots in connections, switches and solar panels.
- **Electroluminescence:** microcracks in photovoltaic cells.

The gain is one of scale: an inspection that would require crews covering thousands of kilometers becomes an image screening problem, with humans reviewing only the flagged cases.

4.5 Data challenges

- **Extreme imbalance:** failures are rare. Overall accuracy is not informative; precision, recall, the PR curve and the cost of false positives (unnecessary visit) versus false negatives (unpredicted failure) are used instead. Precision depends on the prevalence π :

$$\text{precision} = \frac{\pi \text{ TPR}}{\pi \text{ TPR} + (1 - \pi) \text{ FPR}}. \quad (19)$$

With $\pi = 1\%$, a recall (TPR) of 90% and a false positive rate (FPR) of 5%, the precision is 15%: of every 100 alerts, 85 are false.

- **Poor labels:** work orders in free text, causes recorded inconsistently. Natural language processing (today with LLMs) helps to structure the maintenance history.
- **Scattered data:** SCADA, maintenance systems, oil laboratory and inspections in different systems, without a common key.
- **Asset heterogeneity:** different manufacturers, models and ages reduce the volume of comparable data. Transfer learning and semi-supervised models help.

4.6 Product view: Asset health platform (APM)

Problem. Transmission companies, distribution utilities and generators maintain large fleets of assets with maintenance plans based mainly on the calendar. This produces unnecessary maintenance on healthy assets and unexpected failures on degraded assets, with repair costs, lost revenue, continuity compensation and safety risk.

Users and personas.

- **Maintenance engineer:** wants to know which assets need attention and why;
- **Asset manager:** wants to prioritize the maintenance and replacement budget based on risk;
- **Inspection crew:** wants prioritized routes and work orders.

Value proposition. A health and risk index per asset, updated continuously, with the explanation of the signals behind it, integrated with the work order system. The decision moves from “maintain everything every N months” to “maintain what is degrading, when it is degrading”.

MVP scope. Start with **one asset class** for which the data and the value are established:

- **Option A (wind generation):** normal behavior models on turbine SCADA for gearbox, generator and bearings, with alerts and a dashboard per farm;
- **Option B (transmission/distribution):** transformer health index from DGA, loading, age and test history.

In both cases: ranking of assets by risk, explanation of the alerts and recording of the engineer’s feedback (useful or false alert), which becomes a label for the next training cycle.

Evolution (roadmap).

- **V2:** computer vision module for drone inspection (lines, substations, solar panels).
- **V3:** estimation of remaining useful life and optimization of the maintenance plan under budget and crew constraints.
- **V4:** automatic analysis of fault records for fault classification and location, integrated with restoration.

Reference architecture.

- **Ingestion** from SCADA/historians (such as PI), laboratory reports, maintenance management systems and images.
- **Unified asset register** (the key that links the data, in practice the largest task of the project).
- **Models** per asset class, with a common layer of alerts and explainability.
- **Workflow:** alert → engineer's analysis → work order → inspection result → label.

Success metrics.

- **Technical:** alert precision (proportion of confirmed alerts), mean lead time between alert and failure, failure recall.
- **Operational:** reduction of unscheduled failures, reduction of unnecessary maintenance, improvement of DEC/FEC and reduction of compensation.
- **Financial:** maintenance cost per asset and generation availability.

Business model. Subscription per monitored asset (per turbine, per transformer) plus a deployment service. Outcome-based contracts (part of the value tied to the reduction of failures) are possible when there is a defined baseline.

Risks.

- **Alert fatigue:** many false positives lead users to ignore the system. The precision of the alerts weighs more than their number.
- Quality and integration of legacy data.
- Competition from manufacturers (OEMs), which offer monitoring of their own equipment; the differentiation is to be multi-vendor.

5 Distribution and the end consumer

5.1 Context

Distribution is the segment in which the transformation is most visible: MMDG grew after the legal framework for distributed generation (Law 14.300/2022), smart metering is advancing, and new services (time-of-use tariffs, demand response, electric vehicles, a free market for smaller consumers) change the relationship with the consumer. At the same time, old problems, such as non-technical losses, continue to carry a large financial weight for Brazilian distribution utilities.

5.2 Non-technical losses

The problem. The losses of a distribution utility are the difference between the energy injected into the grid and the energy billed. Part of them are *technical losses* (Joule effect in conductors and transformers); the remainder are *non-technical losses*: theft (illegal connections), fraud (meter tampering), and metering and registration errors. In the tariff reviews, ANEEL defines regulatory loss levels that can be passed through to the tariff; whatever exceeds that level is a loss for the utility. In some concession areas, non-technical losses are a relevant fraction of the low-voltage market.

Formulation as classification. For each consumer unit u , the probability of irregularity $p_u = P(\text{irregular} \mid \mathbf{x}_u)$ is estimated from attributes such as:

- **consumption profile:** abrupt drops, zero consumption, low variance, deviation from neighbors with the same profile;
- **history:** previous inspections, confirmed irregularities, meter replacements, complaints;
- **registration:** class, activity, declared load, meter type;
- **spatial context:** irregularity rate in the neighborhood and at the transformer;
- **energy balance:** when there is metering at the transformer,

$$E_k^{\text{NTL}} = E_k^{\text{transformer}} - \sum_{u \in k} E_u^{\text{metered}} - \hat{E}_k^{\text{technical}}, \quad (20)$$

indicates how much is lost under each transformer k and helps to locate where to look. The balance is a small difference between large terms, and its uncertainty adds those of the terms:

$$\sigma^2(E_k^{\text{NTL}}) = \sigma_{\text{transformer}}^2 + \sum_{u \in k} \sigma_u^2 + \sigma_{\text{technical}}^2. \quad (21)$$

The technical loss varies with the square of the current and depends on the shape of the load curve, not only on the energy of the period. The sum over $u \in k$ also assumes that the link between consumer and transformer is correct.

The output is a prioritized list of inspection targets.

Selection bias. The labels come from inspections, and inspections are not random: historically, they were carried out where there was suspicion. The model therefore learns to reproduce the old selection criteria, and not necessarily to identify fraud [29]. In formal terms, training estimates $P(\text{irregular} \mid \mathbf{x}_u, \text{inspected})$, which equals p_u only if, given $\mathbf{x}_u$, the choice of whom to inspect does not depend on the irregularity. Mitigations:

- reserving part of the inspections for random or exploratory sampling (the exploration–exploitation balance);
- learning from positive and unlabeled examples (PU learning), treating the uninspected units as unlabeled and not as regular. The standard form of the method assumes that the labeled positives are a random sample of the positives, which selection by suspicion violates; the applicable variant models the probability of inspection as a function of $\mathbf{x}_u$ and weights the examples by it;
- unsupervised anomaly detection models as a complement.

Metrics. Because inspection capacity is limited, the relevant metrics are the **hit rate at the top of the list** (precision at k) and the **energy recovered per inspection**, and not global metrics such as AUC. The two metrics call for different orderings. Precision at k is maximized by ordering by p_u ; the energy recovered is maximized by ordering by the expected value $p_u \hat{E}_u$, where $\hat{E}_u$ is the estimated recoverable energy of the unit. The second ordering requires a calibrated p_u , because the product uses the value of the probability and not only its rank.

5.3 Load disaggregation (NILM)

Non-intrusive load monitoring estimates the consumption of each appliance from a single aggregate measurement [30]:

$$P(t) = \sum_{i=1}^N p_i(t) + \varepsilon(t). \quad (22)$$

It is an ill-posed source separation problem: many combinations of appliances explain the same aggregate.

Methods.

- **Classical:** event detection (power steps) and factorial hidden Markov models.
- **Deep learning:** networks that map a window of the aggregate to the consumption of one appliance, either the whole window (seq2seq) or the central point (seq2point) [31, 32]. They are trained on public datasets such as REDD and UK-DALE.

Practical limitation. Quality depends strongly on the sampling rate. With meters that record every 15 minutes or 1 hour, only loads with high power and prolonged use can be identified (air conditioning, heater, electric vehicle charger). A high and short load is diluted in the interval mean: a 5 kW electric shower switched on for 5 minutes appears as 1.67 kW in a 15-minute window. Fine disaggregation requires data at a resolution of seconds or finer, obtained with additional devices.

5.4 Distribution grid with MMGD and electric vehicles

- **Low-voltage forecasting:** series per transformer or per consumer are noisy and hard to forecast individually. Probabilistic peak forecasting, including with foundation models, is an approach under evaluation [15].
- **Hosting capacity:** how much MMGD or how many chargers a feeder supports without violating voltage and loading limits. ML surrogates accelerate studies that would require thousands of simulations.
- **Topology and phase identification:** records of which consumer is on which transformer and phase are frequently wrong; the correlation between voltage series from smart meters allows them to be corrected.
- **Detection of unregistered MMGD** from the consumption profile.

5.5 Demand response and efficiency

- **Time-of-use tariffs:** with the Tarifa Branca (the Brazilian time-of-use tariff for low-voltage consumers) and, in the future, more granular tariffs, the consumer needs to understand when to shift consumption. Forecasting and recommendation models personalize this guidance.
- **Control of flexible loads:** air conditioning, water heating, vehicle charging and batteries can be scheduled by model predictive control (MPC) or RL to minimize cost while respecting comfort.
- **Commercial buildings and industry:** consumption forecasting and anomaly detection identify waste (equipment switched on outside working hours, degradation of HVAC systems). The DeepMind case, with a reduction of up to 40% in the cooling energy of data centers, illustrates the potential [4].
- **Customer segmentation:** clustering of load curves for tariff design, efficiency campaigns and offers in the free market.

5.6 Privacy and regulation

High-resolution consumption data reveal habits (schedules, presence, appliances). In Brazil, they are personal data under the LGPD (Law 13.709/2018). This requires a legal basis for processing, minimization, transparency and security. Techniques such as aggregation, federated learning and differential privacy reduce the exposure. The rollout of smart metering in Brazil is also more gradual than in Europe, which limits the availability of high-resolution data at large scale.

5.7 Product view: Loss and grid intelligence for distribution utilities

Problem. Distribution utilities lose revenue to non-technical losses above the regulatory level and carry out inspections with a low hit rate. At the same time, they need to handle an increasingly complex low-voltage grid (MMGD, electric vehicles) with inaccurate records and little observability.

Users and personas.

- **Energy recovery manager:** wants to maximize the energy recovered with the available inspection budget;
- **Field crew:** wants prioritized routes and targets, with low-effort recording of the result;
- **Distribution planning engineer:** wants to know where the grid will saturate with MMGD and new loads.

Value proposition. More energy recovered per inspection, with a prioritized and explained list of targets that improves at each cycle with the results of the inspections themselves. Afterwards, the same database supports the visibility of the low-voltage grid.

MVP scope.

1. Model for prioritizing inspection targets with billing, registration and inspection history data.
2. Explanation per target (why this customer was indicated).
3. Closed loop: the inspection result returns as a label; part of the inspections is reserved for exploration, which corrects the selection bias.
4. Results dashboard: hit rate and energy recovered compared with the utility's current method (A/B test by region or by batch).

Evolution (roadmap).

- **V2:** energy balance per transformer where there is metering, combining location (where the loss occurs) with prioritization (whom to investigate).
- **V3:** correction of topology and phase records, detection of unregistered MMGD, forecasting of transformer loading.
- **V4:** hosting capacity studies and a consumer-facing product (disaggregation, tariff and efficiency recommendations).

Reference architecture.

- **Ingestion** from the commercial system (billing, registration), the work order management system, smart metering where available, and georeferenced grid data.
- **Prioritization engine** with supervised models (gradient boosting is the standard) and an unsupervised anomaly component.
- **Field application** to record inspection results in a structured form.
- **Data governance** in accordance with the LGPD (access control, anonymization in aggregate analyses, record of processing).

Success metrics.

- **Technical:** precision at the top of the list; calibration of the probabilities.
- **Business:** energy recovered (MWh and R\$) per inspection; reduction of the non-technical loss index; cost per productive inspection.
- **Adoption:** percentage of inspections generated by the system; percentage of results recorded.

Business model. Subscription plus a variable component tied to the energy recovered (success fee). The gain is measurable in reais and appears early, which helps the sale. R&D contracts regulated by ANEEL can also fund the grid modules that are still experimental.

Risks.

- Label quality and selection bias; without an exploratory cycle the model becomes self-referential.
- Social and legal sensitivity: the indication of an irregularity needs a fair procedure; the model prioritizes inspections and does not constitute evidence.
- Geographic or socioeconomic bias, which needs to be monitored to avoid an unjustified concentration of inspections on certain groups.
- Compliance with the LGPD and security of consumer data.

6 Comparative summary

Table 3
Comparison of the four areas.

Area	Core techniques	Product	Main customer	Maturity
Forecasting	GBM, LSTM, Transformers, foundation models	Forecasting platform	generators, trading companies, distribution utilities	high
Grid operation	GNN, PINN, safe RL	Operation copilot	transmission companies, distribution utilities, renewable operators	medium/low
Faults and maintenance	CNN, normal behavior models, survival	Asset health platform	generators, transmission companies, distribution utilities	high
Distribution and consumer	GBM, PU learning, NILM, MPC	Loss and grid intelligence	distribution utilities	high (losses), medium (grid)

Patterns common to the four areas.

1. **Data before models:** in all areas, the largest deployment effort is integrating and cleaning data from legacy systems, not training networks.
2. **Physics as an ally:** models that respect physical laws (PINNs, verification by solver, normalizations such as the clear-sky index) tend to generalize better and to be more readily accepted by engineers.
3. **Human in control:** the products that are viable in the short term support decisions instead of making them alone, and they learn from user feedback.
4. **Shadow mode as a sales strategy:** running in parallel and showing the gain with the customer’s own data reduces the perceived risk.
5. **Suggested order of entry:** forecasting and non-technical losses have early and measurable value and serve as the entry point; predictive maintenance comes next; the operation copilot is the bet with the longest horizon and the highest barrier to entry.

References

[1] Machine learning applications in energy systems: current trends, challenges, and research directions. *Energy Informatics*, 2025. <https://energyinformatics.springeropen.com/articles/10.1186/s42162-025-00524-6>

- [2] Trustworthy artificial intelligence in the energy sector: landscape analysis and evaluation framework. arXiv:2412.07782, 2024.
- [3] DeepMind. Machine learning can boost the value of wind energy, 2019. Case summarized at <https://bestpractice.ai/ai-use-cases/case-studies/energy-utilities/deepmind-increases-value-of-wind-power-by-20-by-predicting-supply-36-hours-in-advance>
- [4] DeepMind. DeepMind AI reduces Google data centre cooling bill by 40%, 2016. <https://deepmind.google/blog/deepmind-ai-reduces-google-data-centre-cooling-bill-by-40/>
- [5] ONS. Avaliação e diagnóstico do sistema de previsão da irradiância solar e geração. Workshop presentation. https://www.ons.org.br/AcervoDigitalDocumentosEPublicacoes/Apresentacao_Workshop.pdf
- [6] Desenvolvimento e implantação no ONS de um modelo de previsão de geração eólica. ABEEólica. <https://abeeolica.org.br/wp-content/uploads/2017/07/Art.pdf>
- [7] Previsão horária de geração fotovoltaica utilizando deep learning. *Observatorio de la Economía Latinoamericana*. <https://ojs.observatoriolatinoamericano.com/ojs/index.php/olel/article/view/11654>
- [8] B. Lim, S. Ö. Arık, N. Loeff, T. Pfister. Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 2021.
- [9] A. F. Ansari et al. Chronos: learning the language of time series. *Transactions on Machine Learning Research*, 2024.
- [10] A. Das, W. Kong, R. Sen, Y. Zhou. A decoder-only foundation model for time-series forecasting. *ICML*, 2024.
- [11] Foundation models for clean energy forecasting: a comprehensive review. arXiv:2507.23147, 2025.
- [12] A benchmark for electrical load forecasting across grid levels: time-series transformers outperform established methods. arXiv:2607.15705, 2026.
- [13] Time series foundation models for energy load forecasting on consumer hardware: a multi-dimensional zero-shot benchmark. arXiv:2602.10848, 2026.
- [14] Explainable load forecasting with covariate-informed time series foundation models. arXiv:2604.28149, 2026.
- [15] Probabilistic low-voltage peak load forecasting with time series foundation models evaluated on application-oriented metrics. arXiv:2607.01966, 2026.
- [16] GridFM: a physics-informed foundation model for multi-task energy forecasting using real-time NYISO data. *Energies*, 19(2):357, 2026. <https://doi.org/10.3390/en19020357>
- [17] Graph neural networks for efficient AC power flow prediction in power grids. arXiv:2502.05702, 2025.
- [18] Initial estimate of AC optimal power flow with graph neural networks. *Electric Power Systems Research*, 2024.
- [19] Graph neural network-accelerated network-reconfigured optimal power flow. arXiv:2410.17460, 2024.
- [20] P. L. Donti, D. Rolnick, J. Z. Kolter. DC3: a learning method for optimization with hard constraints. *ICLR*, 2021.
- [21] A. Marot et al. Learning to run a power network challenge for training topology controllers. *Electric Power Systems Research*, 2020.
- [22] Graph reinforcement learning for power grids: a comprehensive survey. *Energy and AI*, 2025.
- [23] Proximal policy optimization with graph neural networks for optimal power flow. arXiv:2212.12470, 2022.
- [24] Optimizing energy management of smart grid using reinforcement learning aided by surrogate models built using physics-informed neural networks. arXiv:2510.17380, 2025.
- [25] A review of safe reinforcement learning methods for modern power systems. arXiv:2407.00304, 2024.
- [26] Towards systematic generalization for power grid optimization problems. arXiv:2605.02026, 2026.

- [27] M. Raissi, P. Perdikaris, G. E. Karniadakis. Physics-informed neural networks. *Journal of Computational Physics*, 378:686–707, 2019.
- [28] Survey on methods for detection, classification and location of faults in power systems using artificial intelligence. arXiv:2507.10011, 2025.
- [29] P. Glauner et al. The challenge of non-technical loss detection using artificial intelligence: a survey. *International Journal of Computational Intelligence Systems*, 2017.
- [30] G. W. Hart. Nonintrusive appliance load monitoring. *Proceedings of the IEEE*, 80(12):1870–1891, 1992.
- [31] J. Kelly, W. Knottenbelt. Neural NILM: deep neural networks applied to energy disaggregation. *ACM BuildSys*, 2015.
- [32] C. Zhang, M. Zhong, Z. Wang, N. Goddard, C. Sutton. Sequence-to-point learning with neural networks for non-intrusive load monitoring. *AAAI*, 2018.